

A380X flyPad failures

The A380X flyPad failure list says what each failure really fails, lists the landing gear sensor and actuator failures, and adds brake, trim air, feed tank pump, crossfeed valve and transfer pump failures, with the trim air duct overhear and fuel pump alerts of the FCOM, and engine flameouts and seizures with the A380 in-flight relight envelope

Aircraft	A380X
Where to find it	flyPad > Failures (ATA 21, 24, 26, 28, 32, 34, 72); overhead FUEL and AIR panels; ENG MASTER levers and ENG START selector; E/WD, STATUS and the SD FUEL, COND and ENGINE pages
Branch	core/failures-a380x, based on master (FlyByWire master 255546cb0 merged in), with core/fire and core/ecam-fws merged in (2026-10-05, for the engine failures: they cut the fuel on the LP valve starvation of core/fire and complete the ENG FAIL ECAM procedures of core/ecam-fws). Merge core/fire and core/ecam-fws first, or with it. It also carries the shared Rust engine failure module of the A32NX engine failures (the fbw-common part of core/failures-a32nx, identical files). The fire and ECAM features are described in the core/fire and core/ecam-fws manuals.
Status	Gear failures tested in the sim by the user: OK (2026-10-03). Brake / trim air failures and feed pump / crossfeed valve failures tested in the sim by the user: OK (2026-10-04); the trim air duct overhear / jammed valve alerts and the transfer pump failures with the fuel follow-ups too (2026-10-04). Clearer names: not sim-tested (labels only, checked by the a380.spec.ts unit tests). Engine flameout / seizure failures: sim-tested OK 2026-10-05 (ground + in-flight, recorded), on develop.
Pictures	No pictures: the failures page and the ECAM alerts are the stock FBW displays.

1. Landing gear sensor and actuator failures listed, ROLLOUT removed
2. Clearer failure names
3. Brake and trim air failures
4. Trim air duct overhear and jammed valve alerts
5. Feed tank pump and crossfeed valve failures
6. Transfer pump failures and fuel follow-ups
7. A380X engine failures (flameout, seizure, relight)

1. Landing gear sensor and actuator failures listed, ROLLOUT removed

Aircraft:	A380X	Where:	flyPad > Failures > ATA 32 Landing Gear	Status:	Tested in the sim by the user: OK (2026-10-03). Unit tests: a380.spec.ts.
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What it is for

The landing gear proximity sensor and jammed actuator failures (ids 32004 to 32025) were already simulated by the common FBW gear system and mapped in the A380X systems, but the flyPad listed only one of them, and with the wrong name: "uplock nose gear #1" failed the **right main gear** uplock sensor #1. All 18 failures are now listed, with the ids and names of the Rust failure map (the same as on the A32NX).

The **ROLLOUT** failure (ATA 22) only set an FCDC input that nothing reads (there is no steering control system yet). It is no longer listed; its id stays in the code so the C++ failure list keeps in step.

What changed in the list

Id	Before	Now
32004	not listed	Proximity sensor damage uplock nose gear #1
32005	not listed	Proximity sensor damage downlock nose gear #2
32006	Proximity sensor damage uplock nose gear #1 (but failed the right gear)	Proximity sensor damage uplock right gear #1
32007	not listed	Proximity sensor damage downlock right gear #2

32008	not listed	Proximity sensor damage uplock left gear #2
32009	not listed	Proximity sensor damage downlock left gear #1
32010 to 32015	not listed	Proximity sensor damage closed / opened nose, right, left gear door #1 / #2 (six entries)
32020	not listed	Nose gear jammed actuator
32021, 32022	not listed	Left / Right main gears (body + wing) jammed actuator
32023	not listed	Nose gear door jammed actuator
32024, 32025	not listed	Left / Right main gear doors jammed actuator
22001	ROLLOUT (ATA 22)	not listed

NOTE

A door *uplock* sensor ("closed ... door") detects the door closed, a door *downlock* sensor ("opened ... door") the door open. The 32021 / 32022 and 32024 / 32025 names come from the clearer names feature below: one failure jams the whole left or right main gear group.

What you should see

Situation	What you should see
A jammed gear actuator failure set, gear lever moved	That gear (or gear door) group stays where it was: the common FBW gear system freezes its actuator.
A proximity sensor failure set	That sensor gives its LGCIS a wrong signal. The gear logic and its indications are the existing FBW gear model: this branch only lists the failures.
ATA 22 list	No ROLLOUT entry.

How to use

1. Open flyPad > Failures > ATA 32 Landing Gear.
2. Pick the sensor or actuator failure by its name (the table above gives what each one fails) and activate it.
3. Deactivate it to clear the failure.

How to test

#	Do this	Expected result
1	Open flyPad > Failures > ATA 32 Landing Gear.	The 12 "Proximity sensor damage ..." and 6 "... jammed actuator" entries are listed, among them "Proximity sensor damage uplock right gear #1" and "Proximity sensor damage uplock nose gear #1".
2	Open flyPad > Failures > ATA 22 Auto Flight.	No ROLLOUT entry (FMC-A, FMC-B, FMC-C, FCU 1, FCU 2 are still there).
3	In flight, gear down, activate Nose gear jammed actuator , then set the gear lever UP.	The nose gear does not retract; the main gears do.
4	Deactivate the failure.	The nose gear retracts.

2. Clearer failure names

Aircraft: A380X | Where: flyPad > Failures | Status: Not sim-tested (labels only). Unit tests: a380.spec.ts.

What it is for

The failure names were audited ATA by ATA against what the systems code fails (the Rust failure map, the C++ failure lists and the TypeScript consumers). Names that were misleading or misspelt now say what is failed. Only the labels change: the ids and the failures themselves are the same.

Old name and new name

ATA	Before	Now
21	Foward Cargo Isolation Valve / Foward Cargo Extraction Fan	Forward Cargo Isolation Valve / Forward Cargo Extraction Fan
24	TR ESS	ESS TR
24	TR APU	APU TR
24	AC 247XP	AC EHA (247XP)
24	DC 247PP	DC EHA (247PP)
24	DC 309PP	DC APU (309PP)
26	Engine 1 Loop A ... Engine 4 Loop B	Engine 1 fire detection loop A ... Engine 4 fire detection loop B
26	APU Loop A / APU Loop B	APU fire detection loop A / APU fire detection loop B
26	Main Landing Gear Bay Loop A / B	MLG bay fire detection loop A / B
32	LGCIU 1 Power supply, LGCIU 2 Power supply	LGCIS 1 power supply, LGCIS 2 power supply
32	LGCIU 1 Internal error, LGCIU 2 Internal error	LGCIS 1 internal error, LGCIS 2 internal error
32	Main left / right gear jammed actuator	Left / Right main gears (body + wing) jammed actuator
32	Main left / right gear door jammed actuator	Left / Right main gear doors jammed actuator
34	RA SYS A Interrupted (B, C)	RA SYS A antenna interrupted (B, C)
34	RA SYS A Direct Coupling (B, C)	RA SYS A antenna direct coupling (B, C)
34	TERR 1 / TERR 2	TERR SYS 1 / TERR SYS 2

Limitations and why

Limitation	Why	What to do
"AC ESS" (24105) and "AC EMER" (24104) keep their names.	The A380X electrical model calls the real AC ESS busbar "AC ESS SHED" and the AC EMER busbar "AC ESS"; the labels already follow the real busbars.	Use "AC ESS" for the AC ESS busbar, "AC EMER" for the AC EMER busbar.
"XPDR 2" (34004) and "NSS AVNCS ANSU 2" (46002) keep their names but fail nothing on this branch.	Nothing reads them on master. They are wired on <code>core/a380x-elec</code> .	Test them with that branch.

How to use

1. Open flyPad > Failures and pick the failure by its new name (table above).

How to test

#	Do this	Expected result
1	Open flyPad > Failures > ATA 24 Electrical Power.	The entries read "ESS TR", "APU TR", "AC EHA (247XP)", "DC EHA (247PP)", "DC APU (309PP)".
2	Open ATA 26 Fire Protection.	The loop failures read "Engine 1 fire detection loop A" ... "MLG bay fire detection loop B".
3	Open ATA 32 Landing Gear and ATA 34 Navigation.	"LGCIS 1 power supply" ... "LGCIS 2 internal error"; "RA SYS A antenna interrupted", "RA SYS A antenna direct coupling", "TERR SYS 1", "TERR SYS 2".
4	Open ATA 21 Air Conditioning.	"Forward Cargo Isolation Valve" and "Forward Cargo Extraction Fan" are spelt right.
5	Activate "ESS TR" with the aircraft powered.	The ESS TR fails, as "TR ESS" did before (same id 24002).

3. Brake and trim air failures

Aircraft: A380X | **Where:** flyPad > Failures > ATA 32 and ATA 21 | **Status:** Tested in the sim by the user: OK (2026-10-04); the brake leak works but shows nothing on the SD HYD page.

What it is for

The A380X hydraulic and air conditioning models already had brake leak and trim air faults, but the flyPad had no failure for them. Eleven failures are now listed (same ids as on the A32NX for the brakes), and a spec checks that every id of the systems failure map is listed with its label.

flyPad name (ATA)	What it fails
Brakes NORM circuit leak (green hydraulic) (32)	The NORM brakes are fed by the GREEN system in this model: GREEN fluid leaks out (0.1 gal/s) while the circuit is pressurised.
Brakes ALTN circuit leak (yellow hydraulic) (32)	The ALTN brakes (with the brake accumulator) are fed by the YELLOW system: YELLOW fluid leaks out while pressurised.
Brakes ALTN accumulator gas leak (32)	The ALTN brake accumulator loses its gas charge: the stored brake pressure drops when the YELLOW system is not pressurised.
Cockpit / Main deck zone 1 / Upper deck zone 1 / Forward cargo trim air valve jammed (21)	The trim air valve of that zone stays where it was when the failure was set.
Cockpit / Main deck zone 1 / Upper deck zone 1 / Forward cargo duct overheat (hot trim air) (21)	200 deg C air is blown into the zone duct while its hot air valve is open. The zone gets too hot and the controller closes the trim air valves; HOT AIR OFF stops the overheat (A380 FCOM COND DUCT OVHT procedure).

Limitations and why

Limitation	Why	What to do
No ECAM alert and no SD indication for the brake failures.	The SD HYD page has no brake leak indication. The trim air failures now have their alerts and SD marks: see "Trim air duct overheat and jammed valve alerts" below.	Watch the hydraulic quantity / pressure and the brake pressure.
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Brakes: flyPad > Failures > ATA 32 Landing Gear, the three "Brakes ..." entries.
2. Trim air: flyPad > Failures > ATA 21 Air Conditioning, the "... trim air valve jammed" and "... duct overheat (hot trim air)" entries.

How to test

#	Do this	Expected result
1	A380X on the ground, four engines running (GREEN and YELLOW pressurised). Activate Brakes NORM circuit leak (green hydraulic) .	The GREEN system loses fluid; YELLOW is not affected. Nothing is shown for the leak itself on the SD HYD page.
2	Deactivate it, activate Brakes ALTN circuit leak (yellow hydraulic) .	The YELLOW system loses fluid; GREEN is not affected.
3	Set the parking brake, then stop the engines (YELLOW no longer pressurised). Activate Brakes ALTN accumulator gas leak .	The brake pressure held by the accumulator drops.
4	In flight, HOT AIR pushbuttons ON, activate Cockpit duct overheat (hot trim air) . Open the SD COND page.	The cockpit zone gets hotter; then the trim air valves close. Setting HOT AIR OFF stops the overheat.
5	Activate Cockpit trim air valve jammed , then change the cockpit temperature selection.	The cockpit trim air valve no longer moves: the cockpit temperature stops following the selection; the other zones still regulate.

4. Trim air duct overheat and jammed valve alerts

Aircraft: A380X | **Where:** overhead AIR panel (HOT AIR 1 / 2, PACK 1 / 2); E/WD, STATUS; SD COND page | **Status:** Tested in the sim by the user: OK (2026-10-04). Rust tests (air_conditioning), unit tests: CondTrimAirFlags.spec.ts, CondTrimAirMarks.spec.ts.

What it is for

The trim air failures above now give the alerts of the A380 FCOM (PRO-ABN-ECAM-10-21-10). The temperature controller monitors each trim air duct and valve: a duct above 70 deg C for 1.2 s is an overheat, latched until its HOT AIR pushbutton is set OFF; a valve that does not follow its demand is jammed. The FWS reads both from the controller and raises:

- **COND DUCT OVHT** (master caution, single chime, FAULT light in the HOT AIR pushbutton, SD COND page) with the CKPT DUCT OVHT, CABIN DUCT OVHT and FWD CARGO DUCT OVHT cases, the deferred procedure **WHEN DUCT OVHT OUT**, and its STATUS lines (INOP SYS, CABIN TEMP REGUL DEGRADED, FWD CRG TEMP REGUL DEGRADED).
- **COND FWD CARGO TEMP REGUL FAULT** for a jammed forward cargo trim air valve: CARGO TRIM AIR VLV FAULT, CARGO TEMP ... MONITOR, STATUS INFO FWD CRG TEMP REGUL DEGRADED (SD COND page only: no chime, no master light).
- SD COND page: the amber OVHT mark of an overheated cockpit or cabin zone duct and the trim air valve mark of a jammed valve (before: never shown).

What you should see

Situation	What you should see
Cockpit duct overheat	COND DUCT OVHT, CKPT DUCT OVHT: HOT AIR 2 ... OFF; if the hot air valve stays open (HOT AIR JAMMED OPEN): PACK 2 ... OFF. HOT AIR 2 FAULT light, OVHT on the SD COND page.

Main deck zone duct overheat	CABIN DUCT OVHT: HOT AIR 1 ... OFF (HOT AIR 2 for an upper deck zone); if jammed open, PACK 1 (2) ... OFF; CARGO TEMP ... MONITOR.
Forward cargo duct overheat	FWD CARGO DUCT OVHT: HOT AIR 1 ... OFF; if jammed open, PACK 1 ... OFF.
Overheat gone (duct cooled, HOT AIR OFF)	Deferred procedure WHEN DUCT OVHT OUT: HOT AIR 1 (2) ... ON, and PACK 1 (2) ... ON if it was switched off.
Forward cargo trim air valve jammed	COND FWD CARGO TEMP REGUL FAULT; STATUS INFO FWD CRG TEMP REGUL DEGRADED; the valve mark on the SD COND page.

Limitations and why

Limitation	Why	What to do
Which HOT AIR valve supplies which zone is a design choice.	The FCOM does not say it: the main deck zones and the forward cargo are on HOT AIR 1, the cockpit and the upper deck on HOT AIR 2.	-
A jammed forward cargo valve counts as a degraded regulation (no INOP SYS).	Design choice: the aircraft has two forward cargo trim air valves and the FCOM says a jammed valve loses the "optimized" regulation.	-
The alerts are inhibited during the takeoff and the landing (FWS phases 3 to 7, 9, 10).	The FCOM gives the inhibition as a figure: that of COND HOT AIR FAULT / FWD CARGO ISOL FAULT is used.	Test on the ground before takeoff or in cruise.
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 21 Air Conditioning and activate a "... duct overheat (hot trim air)" or a "... trim air valve jammed" failure.
2. Follow the ECAM with the HOT AIR and PACK pushbuttons of the overhead AIR panel; check the SD COND page.
3. Deactivate the failure; when the duct has cooled, follow WHEN DUCT OVHT OUT.

How to test

#	Do this	Expected result
1	A380X in cruise (or on the ground before takeoff), packs and HOT AIR 1 / 2 ON. Activate Cockpit duct overheat (hot trim air) .	After the duct passes 70 deg C: master caution, E/WD COND DUCT OVHT, CKPT DUCT OVHT with HOT AIR 2 ... OFF; HOT AIR 2 FAULT light; SD COND page with the amber OVHT mark on the cockpit.
2	Set HOT AIR 2 OFF. Open the STATUS page.	The line is done; STATUS INOP SYS and INFO lines of COND DUCT OVHT.
3	Deactivate the failure and wait for the duct to cool.	The alert goes; WHEN DUCT OVHT OUT asks HOT AIR 2 ... ON.
4	Activate Main deck zone 1 duct overheat (hot trim air) .	COND DUCT OVHT, CABIN DUCT OVHT with HOT AIR 1 ... OFF and CARGO TEMP ... MONITOR.
5	Deactivate it. Activate Forward cargo trim air valve jammed , then change the cargo temperature selection.	COND FWD CARGO TEMP REGUL FAULT (no chime): CARGO TRIM AIR VLV FAULT, CARGO TEMP ... MONITOR; STATUS INFO FWD CRG TEMP REGUL DEGRADED.

6	Activate Cockpit trim air valve jammed and change the cockpit temperature selection.	The trim air valve mark appears on the cockpit zone of the SD COND page.
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5. Feed tank pump and crossfeed valve failures

Aircraft: A380X | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL | **Status:** Tested in the sim by the user: OK (2026-10-04), after the fix that holds the MSFS pump / valve switch.

What it is for

Twelve new failures: the main and standby pump of each feed tank and the four crossfeed valves. The FUEL panel pushbuttons now set a selection, and the systems host moves the MSFS pump or valve switch from it: a failed pump is held off, a jammed valve is held where it was. The FAULT lights, the SD FUEL page and the FCOM alerts compare the selection with the real state (A380 FCOM PRO-ABN-ECAM-10-28, DSC-28-20).

flyPad name (ATA 28)	What it fails
Feed tank 1 main pump ... Feed tank 4 main pump	That main pump stops (held off).
Feed tank 1 standby pump ... Feed tank 4 standby pump	That standby pump stops (held off).
Crossfeed valve 1 jammed in place ... Crossfeed valve 4 jammed in place	That crossfeed valve stays where it was.

What you should see

Situation	What you should see
Feed pump failed, its pushbutton ON	FAULT light in the pushbutton; the pump amber on the SD FUEL page.
One main pump failed	FUEL FEED TK 1 MAIN PMP FAULT (tank 1 to 4), SD FUEL page; procedure FEED TK 1 MAIN PMP ... OFF; STATUS INOP SYS FEED TK 1 MAIN PMP.
One standby pump failed	FUEL FEED TK 1 STBY PMP FAULT , crew awareness; STATUS MORE: REDUND LOSS FEED TK 1 STBY PMP.
Main and standby pumps of one tank failed	FUEL FEED TK 1 MAIN + STBY PMPs FAULT (replaces the two single alerts). After takeoff: CROSSFEED 1 ON, CROSSFEED 2 ON (CROSSFEED 3 and 4 for tanks 3 and 4), FEED TK 1 MAIN PMP OFF, FEED TK 1 STBY PMP OFF, AVOID NEG G LOAD; before takeoff only the two pump lines. STATUS: INOP SYS FEED TK 1 PMPs, FEED TK 1 BY GRVTV ONLY.
Crossfeed valve jammed and its pushbutton moved	The valve disagrees with its pushbutton: FAULT light, valve amber on the SD FUEL page, FUEL CROSSFEED VLV 1 FAULT (crew awareness), STATUS INOP SYS CROSSFEED VLV 1; if the valve is stuck closed and engine 1 is not running, FEED TK 1 NOT USABLE.

Limitations and why

Limitation	Why	What to do
The alerts are inhibited during the takeoff and the landing.	The FCOM gives the inhibition as a figure: the one of JETTISON VLV NOT CLOSED is used (FWS flight phases 4 to 7, 9 and 10).	Test on the ground before takeoff or in cruise.
A pump turned OFF by the crew also gives the FAULT alert.	FCOM: a pump is faulty when it runs at low pressure, is abnormally not running, or is turned off by the crew (except in flight phases 1 and 12).	Expected behaviour.
The gravity feed ceiling (MAX FL 280, 50/MEA) is not simulated.	No gravity feed model.	None.

A full MSFS restart is needed after an update.

The cockpit behaviour XML changes.

Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate a pump or crossfeed valve failure.
2. Follow the ECAM procedure with the overhead FUEL panel pushbuttons; check the SD FUEL page.
3. Deactivate the failure to restore the pump or valve.

How to test

#	Do this	Expected result
1	A380X on the ground, engines running, all feed pumps ON, crossfeed valves in their normal position. Open the SD FUEL page.	No FAULT light, no fuel alert.
2	Activate Feed tank 1 main pump .	FAULT light in the FEED TK 1 MAIN pushbutton, pump amber on the SD, E/WD: FUEL FEED TK 1 MAIN PMP FAULT with FEED TK 1 MAIN PMP ... OFF.
3	Set the FEED TK 1 MAIN pushbutton OFF. Open the STATUS page.	The procedure line is done; STATUS INOP SYS: FEED TK 1 MAIN PMP.
4	Set it ON again and also activate Feed tank 1 standby pump .	The single alerts are replaced by FUEL FEED TK 1 MAIN + STBY PMPs FAULT (on the ground: the two pump OFF lines); STATUS: FEED TK 1 PMPs, FEED TK 1 BY GRVTY ONLY.
5	Deactivate both. Activate Feed tank 2 standby pump only.	FUEL FEED TK 2 STBY PMP FAULT; STATUS MORE: REDUND LOSS FEED TK 2 STBY PMP.
6	Deactivate it. Activate Crossfeed valve 1 jammed in place , then move the CROSSFEED 1 pushbutton.	The valve does not move: FAULT light, valve amber on the SD, FUEL CROSSFEED VLV 1 FAULT, STATUS INOP SYS CROSSFEED VLV 1.
7	Deactivate the failure.	The valve follows its pushbutton again; the alert clears.

6. Transfer pump failures and fuel follow-ups

Aircraft: A380X | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; E/WD, STATUS; SD FUEL page | **Status:** Tested in the sim by the user: OK (2026-10-04). Unit tests: FuelPumpValveFlags.spec.ts, FuelPumpsAndValves.spec.ts, failable-fuel-switches.spec.ts.

What it is for

Eight new failures of the transfer pumps that the FBW fuel transfers use: **L outer tank pump, R outer tank pump, L mid tank fwd pump, R mid tank fwd pump, L inner tank fwd pump, R inner tank fwd pump, Trim tank L pump, Trim tank R pump**. A failed pump is held off whatever the fuel system commands. The pushbutton FAULT light, the amber pump on the SD FUEL page and the FUEL ... PMP FAULT alerts of the A380 FCOM (PRO-ABN-ECAM-10-28) come as soon as a pump fails, runs in an empty tank with its pushbutton ON, or is switched OFF by the crew (except in flight phases 1 and 12). A repaired pump follows the automatic transfers again.

- **FUEL FEED TK 1(2)(3)(4) LEVEL LO** now triggers for a single low feed tank: the four tanks shared one 30 s confirmation, so only all four low together could trigger it.
- The **CROSSFEED** pushbutton OPEN light now follows its valve.

- The sensed CROSSFEED ... ON and ALL FEED TKs PMPs ON items of the ECAM procedures tick on the pushbuttons, even with a jammed valve or a failed pump.
- Corrupted degree signs in the ECAM texts are fixed.

What you should see

Situation	What you should see
L outer tank pump failed	FAULT light; FUEL L OUTR TK PMP FAULT : L OUTR TK PMP ... OFF; STATUS: OUTR TKs XFR BY GRVTY ONLY, INOP SYS L OUTR TK PMP.
L inner (mid) tank fwd pump failed	FUEL L INR (MID) TK FWD PMP FAULT : L INR (MID) TK FWD PMP ... OFF; STATUS INOP SYS L INR (MID) TK FWD PMP.
Its aft pump switched OFF too	FUEL L INR (MID) TK FWD+AFT PMPs FAULT replaces the single alerts: both pumps ... OFF; L INR (MID) TK NOT USABLE; INOP SYS L INR (MID) TK PMPs.
Trim tank L and R pumps failed	FUEL TRIM TK L+R PMPs FAULT : both pumps ... OFF, PITCH ATT < 3° FOR GRVTY FWD XFR, TRIM TK FEED ... OPEN, TRIM TK : 2300 KG MAX NOT USABLE; STATUS: TRIM TK XFR BY GRVTY ONLY, TRIM TK GRVTY XFR : AROUND 3000 KG / HR, INOP SYS TRIM TK PMPs.
One feed tank low	FUEL FEED TK n LEVEL LO for that tank after 30 s.
CROSSFEED pushbutton ON	The OPEN light comes on when the valve is open.

Limitations and why

Limitation	Why	What to do
The aft gallery pumps (inner and mid tank aft) cannot fail.	They never run in the simulation; switching them OFF still gives their alert.	-
No gravity transfer.	Not simulated: the STATUS gravity lines are information only.	None.
The alerts are inhibited during the takeoff and the landing.	Same inhibition as the feed pump alerts above.	Test on the ground before takeoff or in cruise.
A full MSFS restart is needed after an update.	The cockpit behaviour XML changes (OPEN light).	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate one of the eight transfer pump failures.
2. Follow the ECAM with the overhead FUEL panel; check the SD FUEL page.
3. Deactivate the failure to restore the pump; it follows the automatic transfers again.

How to test

#	Do this	Expected result
1	A380X on the ground before takeoff (or in cruise) with fuel in the outer tanks, all fuel pumps ON. Activate L outer tank pump .	FAULT light in the L OUTR TK pushbutton, pump amber on the SD FUEL page, master caution, E/WD FUEL L OUTR TK PMP FAULT with L OUTR TK PMP ... OFF.
2	Set the L OUTR TK pushbutton OFF. Open STATUS.	The line is done; STATUS: OUTR TKs XFR BY GRVTY ONLY, INOP SYS L OUTR TK PMP.
3	Deactivate it and set the pushbutton ON. Activate L inner tank fwd pump , then set the L INR TK AFT pushbutton OFF.	FUEL L INR TK FWD+AFT PMPs FAULT replaces the single alert; L INR TK NOT USABLE.

4	Restore everything. Activate Trim tank L pump and Trim tank R pump .	FUEL TRIM TK L+R PMPs FAULT with its procedure; the degree sign of PITCH ATT < 3° is shown correctly.
5	Restore everything. Set a CROSSFEED pushbutton ON.	Its OPEN light comes on with the valve.

7. A380X engine failures (flameout, seizure, relight)

Aircraft: A380X | **Where:** flyPad > Failures > ATA 72 Engine; pedestal ENG MASTER 1-4 and ENG START selector; overhead AIR panel X BLEED selector; E/WD, STATUS, SD ENGINE page | **Status:** Sim-tested OK 2026-10-05 (ground + in-flight, recorded), on develop: ground card T1-T3 and in-flight card T1-T3 (windmilling, quick and crossbleed relights without APU), earlier T1, T7, T8 (relight with APU bleed) and T9. Not re-tested: two engines out relight at 255 kt, all four engines out. Rust tests (a380_systems engine_failure.rs, pneumatic.rs, hydraulic), native FADEC test (relight_start_test.cpp), unit tests EngineFailAlerts.spec.ts, a380.spec.ts. Not sim-tested on this branch.

What it is for

The A380X port of the A32NX engine failures. Eight new failures in the flyPad chapter ATA 72 Engine. A **flameout** cuts the engine fuel once: the crew relights the engine with ENG MASTER OFF then ON inside the in-flight relight envelope of the A380 FCOM. A **seizure** also stops the core: no relight while the failure is active. The FWS gives the FCOM ENG FAIL, ENG SHUT DOWN, TWO ENGS OUT and ALL ENG FLAME OUT alerts for the four engines.

Id	flyPad failure (ATA 72 Engine)	What the code does
72000-72003	Engine 1 (2)(3)(4) flameout (crew relight possible)	Cuts the engine fuel (latched, MSFS fuel valve 60-63). The engine runs down and windmills. ENG MASTER OFF then ON inside the envelope relights it; it then runs normally while the failure stays listed. Deactivate and activate it again for a new flameout.
72010-72013	Engine 1 (2)(3)(4) seizure (no relight)	Cuts the fuel and stops the core (N3). No relight while the failure is active; once it is cleared the engine relights by the normal procedure.

FCOM references

A380 FCOM (a380_fcom.txt)	What it says	Used for
PRO-ABN-ECAM-10-70 ENG RELIGHT IN FLIGHT, I.174864-175027, chart PDF p.5854	Single engine: "MAX GUARANTEED ALTITUDE : 30000 FT", "MIN SPEED FOR WINDML RELIGHT : 260 KT"; multiple engines: 28 000 ft and 250 kt; ENG START SEL IGN START; "Engine must relight within 30 s after fuel flow increases".	The relight envelope and the 30 s attempt
DSC-70-30 IN FLIGHT, I.113552-113575	Engine start valve opens "if N2 is below 11 %, or if the aircraft airspeed (CAS) is below 260 kt"; it closes above 58.4 %.	The start valve in flight and for a restart
DSC-70-80-30-20, I.112539-112571	Quick relight: MASTER ON-OFF-ON within 30 s, "provided that N2 is greater than 45 %"; starter assisted below 11 % N2, windmilling at or above 260 kt; during a windmilling start the FADEC disconnects both hydraulic pumps.	Quick relight, start type, engine pumps
ENG 1(2)(3)(4) FAIL, I.171712-171981	Triggered when the engine is not running; "If more than two engines failed: ENG (AFFECTED) RELIGHT PROC ... APPLY" (I.171871).	ENG FAIL and RELIGHT PROC APPLY
ENG 1(2)(3)(4) SHUT DOWN, I.172693-172806	Shown when the ENG MASTER is OFF or the ENG FIRE pb pushed; STATUS INOP SYS BTM (engine 2 or 3), CAT 3 DUAL and CAT 3 SINGLE ONLY if the APU is off.	ENG SHUT DOWN STATUS
ENG TWO ENGS OUT ON SAME / OPPOSITE SIDE, I.175216-175635	Two engines failed on the same side / on opposite sides, LAND ANSA, STATUS (CAT 2 / CAT 3 INOP, CAT 2 ONLY).	The two cautions

ENG ALL ENG FLAME OUT, I.114538	The title of the all-engines procedure.	Title of the old ALL ENGINES FAILURE
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The relight envelope, read off the chart (IAS, flight level):

Zone	Where	Needs
3, windmilling start	260-340 kt up to FL300 (multiple engines out: from 250 kt, up to FL280)	ENG START IGN START, ENG MASTER OFF then ON; the start valve stays closed
1 and 2, starter assisted relight	150-260 kt up to FL200; 185-260 kt from FL200 to FL250	ENG START IGN START, ENG MASTER OFF then ON, and starter air: APU BLEED, or X BLEED OPEN with the other engines
2 (and 3), quick relight	220-260 kt up to FL200 (and zone 3), N3 above 45 %	ENG MASTER OFF then ON within 30 s, at any ENG START position

What you should see

Situation	What you should see
Flameout in flight	N3, EGT and FF run down, FF 0, N1 / N3 settle at the windmill values. Amber ENG n FAIL with its FCOM procedure lines (THR LEVER n IDLE reads its own lever; ENG n RELIGHT PROC CONSIDER).
ENG MASTER OFF in flight	ENG n FAIL goes, ENG n SHUT DOWN comes with its STATUS: INOP SYS BTV for engine 2 or 3, CAT 3 DUAL and CAT 3 SINGLE ONLY with the APU off; the engine 4 bleed INOP comes from engine 4 (it came from another engine before).
Windmilling relight (zone 3, no APU)	At the MASTER ON the fuel comes back; engine at idle in about 35 s; the start valve stays closed; both engine pumps of that engine are off-loaded during the start.
Crossbleed relight (zone 1, X BLEED OPEN)	The start valve opens and the starter is pressurised in about 0.5 s; engine at idle in about 35 s; the start valve closes at N3 about 58 %.
Quick relight (ENG START at NORM)	MASTER OFF then ON at once: the engine relights from about 75 % N3, at idle in about 25 s.
Two engines failed	ENG TWO ENGS OUT ON SAME SIDE or ON OPPOSITE SIDE caution with LAND ANSA; the multiple-engine envelope applies.
Three engines failed	ENG FAIL shows RELIGHT PROC APPLY instead of CONSIDER.
All four engines failed	Red ENG ALL ENG FLAME OUT (the FCOM title; it was ALL ENGINES FAILURE).
On the ground	A failed engine starts again as a normal start, with APU air (ENG START IGN START, MASTER OFF then ON). Two engines shut down for taxi give no TWO ENGS OUT caution.

Design choices

- The envelope governs **every** in-flight relight, also after a crew shutdown without any failure: MSFS would otherwise relight anywhere.
- The quick relight also works in zone 3, where the windmill alone lights the engine.
- A windmilling engine keeps $N3 = 11 \% \times IAS / 260 \text{ kt}$ and $N1 = 1.5 \times N3$ (the FCOM gives no values).
- FBW models the Trent 972 (three spools): the FCOM N2 (GP7200 core) is read on the Trent N3. ENG FAIL keeps the $N3 < 50 \%$ threshold of FBW (the FCOM says N2 below 61 %).

- TWO ENGS OUT is shown only with exactly two engines out (with three, ENG FAIL RELIGHT PROC APPLY; with four, ALL ENG FLAME OUT).
- During a windmilling start the two engine pumps are off-loaded (depressurised), not disconnected: the ENG PUMPS DISC clutch cannot be reconnected in flight.
- Once a relight is granted in flight the FADEC holds the MSFS core at 25 % N2: MSFS only burns above 20 % N2, and its starter only turns with the APU bleed, so without it a windmilling or crossbleed relight would never burn.

Limitations and why

Limitation	Why	What to do
ENG TWO ENGS OUT ON OPPOSITE SIDE has no PACK line.	Not added yet (FCOM I.175519).	-
Some TWO ENGS OUT STATUS lines are missing.	Only CAT 2 / CAT 3 INOP and CAT 2 ONLY are written; the other STATUS lines of I.175434-175632 are not.	-
The SD ENGINE page does not show the igniters.	The A380X SD page has no igniter indication yet.	-
L:A32NX_PNEU_ENG_n_STARTER_PRESSURIZED stays 1 after the start valve closes.	The duct keeps its pressure; the indication follows the duct.	-
Two engines out relight at 255 kt and all four engines out not re-tested in the sim.	Covered by the Rust tests only.	-
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 72 Engine and activate a flameout or a seizure.
2. Follow the ECAM. To relight: get inside the envelope (table above), ENG START selector (pedestal, engine panel) IGN START, ENG MASTER OFF for 30 s, then ON. Below 260 kt without the APU, set the X BLEED selector (overhead AIR panel) to OPEN for starter air.
3. For a quick relight, set the ENG MASTER OFF then ON at once (within 30 s, N3 above 45 %), at any ENG START position.
4. When the engine is at idle: ENG START NORM, X BLEED AUTO; clear the failure in the flyPad (a seizure must be cleared before a relight).

How to test

Short card in the sim-test format (from `reports/test-card-b1-inflight.md` and `test-card-b1-ground-2026-10-05.md`). Needs one MSFS restart after installing the branch.

T1 Windmilling relight: engine 1 relights by the airflow alone, no APU

Setup: in flight at FL200, 300 kt, AP and A/THR ON, APU OFF, X BLEED selector AUTO (overhead AIR panel), ENG START selector NORM (pedestal, engine panel with the four ENG MASTER levers).

#	Do this	Expected result
1	flyPad > Failures > ATA 72 > "Engine 1 flameout (crew relight possible)": activate it.	Amber ENG 1 FAIL; engine 1 FF 0, N3 and EGT fall.
2	ENG START selector: IGN START. ENG MASTER 1: OFF, wait 30 s.	ENG 1 SHUT DOWN replaces ENG 1 FAIL.

3	ENG MASTER 1: ON.	FF appears within a few seconds, engine 1 at idle within about 60 s, the alerts clear; SD ENGINE page: the engine 1 start valve stays closed.
4	ENG START selector NORM; flyPad: deactivate the failure.	-

T2 Quick relight: engine 4 relights at NORM

Setup: FL150, 240 kt, ENG START selector NORM.

#	Do this	Expected result
1	Activate "Engine 4 flameout (crew relight possible)"; at once (within 5 s) ENG MASTER 4 OFF, then ON.	Engine 4 FF and EGT rise within a few seconds, back at idle, ENG 4 FAIL goes; the selector stays at NORM.
2	flyPad: deactivate the failure.	-

T3 Starter assisted relight with crossbleed air: engine 3

Setup: FL150, 210 kt, APU OFF.

#	Do this	Expected result
1	X BLEED selector OPEN. Activate "Engine 3 flameout (crew relight possible)".	ENG 3 FAIL.
2	ENG START selector IGN START; ENG MASTER 3 OFF, wait 30 s; ENG MASTER 3 ON.	SD ENGINE page: engine 3 start valve open, FF within a few seconds, engine 3 at idle within about 60 s, start valve closed again, ENG 3 FAIL goes.
3	ENG START NORM, X BLEED AUTO, deactivate the failure.	-

T4 On the ground: relight with APU air, and two engines out for taxi

Setup: on the ground, four engines at idle, parking brake ON, APU AVAIL, APU BLEED pb ON (overhead AIR panel), ENG START NORM.

#	Do this	Expected result
1	Activate "Engine 1 flameout"; when N1 is below 5 %: ENG MASTER 1 OFF, ENG START IGN START, wait 30 s, ENG MASTER 1 ON.	ENG 1 FAIL, then a normal start: start valve open, engine 1 at idle within about 60 s, ENG 1 FAIL goes.
2	ENG START NORM, deactivate the failure. ENG MASTER 1 OFF and ENG MASTER 4 OFF, wait 20 s.	No TWO ENGS OUT ON OPPOSITE SIDE caution, no ENG 1 / ENG 4 FAIL.

NOTE

Recording used for the 2026-10-05 tests: per engine FUEL_CUT, RELIGHT_ATTEMPT, RELIGHT_IGNITION, ENGINE_STATE, ENGINE_N3, PNEU starter valve / pressurized, MSFS TURB ENG N2, GENERAL ENG COMBUSTION and fuel flow, with the airspeed, altitude, ENG START selector, X BLEED and APU BLEED.